

FUNCTIONAL SPECIFICATION DOCUMENT (FSD)

Rising Waters – AI-Based Flood Prediction System

Project Documentation

Project Name: Rising Waters – AI-Based Flood Prediction System

Team ID: RW-2026-01

Project Duration: 24 June 2026 – 20 July 2026

Department: Electronics and Communication Engineering

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1. Document Purpose:

This Functional Specification Document (FSD) defines the functional and technical specifications of the **Rising Waters – AI-Based Flood Prediction System**. The document serves as a comprehensive reference for understanding the objectives, architecture, functionalities, workflows, and implementation details of the application. It enables faculty members, evaluators, and developers to understand the complete system without referring to the source code.

The document also establishes a standardized description of system behavior, ensuring that all stakeholders share a common understanding of the project's scope and functionality. Furthermore, it acts as a reference for future maintenance, enhancements, testing, and deployment activities.

2. Application Summary:

Floods are among the most destructive natural disasters, causing extensive damage to human lives, agriculture, infrastructure, and the economy. Traditional flood forecasting methods rely heavily on manual observations, historical records, and delayed weather reports, limiting their ability to provide timely and accurate warnings.

Rising Waters is an Artificial Intelligence-based Flood Prediction System developed to provide early flood risk assessment using Machine Learning techniques. The application utilizes the **Extreme Gradient Boosting (XGBoost)** algorithm to analyze environmental and hydrological parameters and predict the probability of flooding.

The system is developed using the **Flask** framework for backend processing and **HTML, CSS, and JavaScript** for the frontend interface. A **MySQL** database stores prediction records and user information, enabling efficient data management and future analysis.

The application assists disaster management authorities, researchers, farmers, and citizens by providing accurate predictions that support early warning systems and informed decision-making.

Major Features

- AI-based Flood Prediction
- XGBoost Machine Learning Model
- Responsive Flask Web Application
- Secure MySQL Database
- Historical Prediction Storage
- User-friendly Interface
- Fast Prediction Processing
- Future-ready Architecture

3. Functional Modules:

Module	Description
User Interface	Collects environmental parameters through a responsive web interface.
Data Validation Module	Validates user inputs to ensure correctness and completeness.
Data Processing Module	Cleans, formats, and prepares input data for prediction.
Machine Learning Module	Uses the trained XGBoost model to estimate flood probability.
Prediction Engine	Generates flood risk predictions based on processed data.
Database Module	Stores prediction records and maintains historical data using MySQL.
Results Dashboard	Displays prediction outcomes and historical reports to users.
Administrative Module	Supports system maintenance, monitoring, and future expansion.

4. Technology Stack:

Layer	Technology Used
Programming Language	Python
Frontend	HTML5, CSS3, JavaScript
Backend Framework	Flask
Database	MySQL
Machine Learning	XGBoost
Dataset	Kaggle Flood Prediction Dataset
Version Control	Git & GitHub
Development Environment	Visual Studio Code

5. Functional Requirements

FR-01: User Input

The system shall allow users to enter environmental parameters required for flood prediction.

FR-02: Input Validation

The application shall validate all input values before processing them through the prediction model.

FR-03: Prediction Generation

The XGBoost model shall analyze the submitted data and generate a flood risk prediction with high accuracy.

FR-04: Result Display

The prediction result shall be displayed immediately through an interactive user interface.

FR-05: Data Storage

The application shall store all prediction records in the MySQL database for future reference.

FR-06: Prediction History

Users shall be able to review previously generated prediction records.

FR-07: Secure Processing

The application shall process user requests securely while protecting stored data from unauthorized access.

6. User Roles: Administrator

Responsibilities:

- Monitor system performance
 - Manage database records
 - Update machine learning model
 - Maintain application security
 - Review prediction history
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End User

Responsibilities:

- Enter environmental parameters
 - Generate flood predictions
 - View prediction results
 - Access prediction history
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7. Assumptions:

The successful operation of the system depends on the following assumptions:

- The environmental data entered by users is accurate.
 - The XGBoost model has been trained using quality historical flood data.
 - The MySQL database remains available during system operation.
 - Users have access to a stable internet connection.
 - The Flask server is continuously operational.
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8. Input Field Specification:

The Flood Prediction System accepts environmental and hydrological parameters from users. These inputs are validated before being processed by the XGBoost model.

Field Name	Data Type	Mandatory	Description
Rainfall	Float	Yes	Rainfall measured in millimeters (mm).
River Water Level	Float	Yes	Current river water level in meters.
Temperature	Float	Yes	Ambient temperature in degrees Celsius (°C).
Humidity	Float	Yes	Atmospheric humidity percentage.
Soil Moisture	Float	Yes	Soil moisture content percentage.
Drainage Capacity	Float	Yes	Efficiency of drainage infrastructure.
Previous Flood History	Integer	Yes	Indicates previous flood occurrences (0 = No, 1 = Yes).
Prediction Result	String	System Generated	Displays Low, Moderate, or High Flood Risk.
Prediction Date	Date & Time	System Generated	Timestamp of prediction generation.

9. Database Specification:

The application uses a MySQL database to store prediction records, ensuring efficient retrieval and future analysis.

Column Name	Data Type	Description
Prediction_ID	INT (Primary Key)	Unique prediction identifier.
Rainfall	FLOAT	Rainfall entered by the user.
River_Level	FLOAT	River water level.
Temperature	FLOAT	Temperature value.
Humidity	FLOAT	Humidity percentage.
Soil_Moisture	FLOAT	Soil moisture percentage.
Drainage_Capacity	FLOAT	Drainage efficiency value.
Previous_Flood	INT	Previous flood occurrence.
Prediction	VARCHAR(20)	Flood risk category.
Prediction_Time	DATETIME	Time of prediction.

Database Features:

- Secure storage of prediction history.
- Fast retrieval of previous records.
- Efficient indexing for improved performance.
- Supports future analytical reporting.
- Easy integration with dashboards.

10. Process Flow Summary:

The application follows a systematic workflow to generate flood predictions.

Step 1 – User Access

The user opens the Rising Waters web application through a web browser.

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Step 2 – Data Entry

Environmental parameters such as rainfall, river level, temperature, humidity, and soil moisture are entered through the prediction form.

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Step 3 – Input Validation

The Flask backend validates all input values for completeness and correctness.

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Step 4 – Data Processing

Validated data is preprocessed and converted into a format compatible with the trained XGBoost model.

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Step 5 – Flood Prediction

The XGBoost algorithm evaluates the input parameters and predicts the probability of flooding.

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Step 6 – Result Generation

The application displays one of the following results:

- Low Flood Risk
- Moderate Flood Risk
- High Flood Risk

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Step 7 – Database Storage

Prediction details are automatically stored in the MySQL database for future reference.

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Step 8 – User Review

Users can review prediction results and historical records through the application dashboard.

11. Non-Functional Requirements:

To ensure reliability and usability, the application satisfies the following non-functional requirements:

Performance

- Prediction results should be generated within a few seconds.
- The system should efficiently handle multiple prediction requests.

Reliability

- The application should provide consistent prediction results.
- Data integrity must be maintained throughout system operations.

Security

- User input must be validated to prevent invalid data.
- Prediction records must be stored securely.
- Access to database operations should be controlled.

Usability

- The interface should be intuitive and easy to navigate.
- Users should require minimal technical knowledge to operate the system.

Scalability

- The system should support additional environmental parameters in future versions.
- It should be capable of handling increased user traffic without significant performance degradation.

12. Testing Summary:

The application underwent multiple testing phases to ensure functional correctness and system stability.

Test Type	Objective	Status
Unit Testing	Verify individual modules	Passed
Functional Testing	Validate application features	Passed
Database Testing	Confirm data storage and retrieval	Passed
Machine Learning Testing	Verify prediction accuracy	Passed
Integration Testing	Validate interaction between modules	Passed
User Interface Testing	Ensure responsiveness and usability	Passed

Test Outcomes

- Successful prediction generation.
- Accurate database storage.
- Responsive user interface.

- Stable integration between Flask, MySQL, and XGBoost.
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13. Out of Scope:

The current implementation does not include the following features:

- Real-time weather API integration.
- IoT sensor-based environmental monitoring.
- SMS and email alert notifications.
- GIS-based flood mapping.
- Mobile application support.
- Multi-language interface.
- Cloud-based deployment with distributed infrastructure.

These features are planned for future versions of the project.

14. Future Enhancements:

The Rising Waters system has significant potential for future expansion.

Planned enhancements include:

- Integration with live weather APIs for real-time environmental data.
 - GPS-based flood location tracking.
 - Interactive GIS maps for flood visualization.
 - SMS, email, and mobile push notifications for emergency alerts.
 - Android and iOS mobile application development.
 - IoT sensor integration for automated data collection.
 - Satellite imagery analysis using deep learning techniques.
 - Multi-language support for wider accessibility.
 - Cloud deployment for high availability and scalability.
 - Predictive analytics dashboard for government agencies.
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15. Conclusion:

Rising Waters – AI-Based Flood Prediction System demonstrates the effective use of Artificial Intelligence and Machine Learning in disaster management. By integrating the XGBoost algorithm with Flask and MySQL, the application provides accurate flood risk predictions through a simple and user-friendly web interface.

The project successfully addresses the limitations of traditional flood forecasting methods by offering data-driven decision support, enabling early warning and improved preparedness. The modular architecture ensures maintainability, scalability, and adaptability for future enhancements.

The implementation of this system contributes toward improving public safety, supporting emergency response teams, protecting infrastructure, and minimizing economic losses caused by floods. With future integration of real-time weather data, IoT devices, and GIS technologies, Rising Waters has the potential to evolve into a comprehensive intelligent flood management platform.

16. Sign-off:

This Functional Specification Document represents the finalized functional and technical requirements of the **Rising Waters – AI-Based Flood Prediction System**.

It serves as the official reference for project implementation, testing, evaluation, and future enhancements.

Project Name: Rising Waters – AI-Based Flood Prediction System

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